CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of digits in $2^6 \times 5^9$ is: 1

(a) 8 **(b)** 10 **(c)** 11 **(d)** 9

2. The ones digit of 13^{42} is: 1

(a) 3 **(b)** 1 **(c)** 5 **(d)** 9

3. If $2^x = 512$, then 2^{x-4} equals: 1

(a) 32 **(b)** 16 **(c)** 508 **(d)** 8192

4. The value of $(3^{-1} + 6^{-1})^{-1}$ is: 1

(a) 2 **(b)** $\frac{1}{2}$ **(c)** 18 **(d)** $\frac{1}{9}$

5. Which of the following numbers is the greatest? 1

(a) 7.5×10^{-3} **(b)** 0.9×10^{-2} **(c)** 0.06×10^{-1} **(d)** 85×10^{-4}

6. If $3^x = 4$, then 3^{x+2} equals: 1

(a) 13 **(b)** 8 **(c)** 36 **(d)** 729

7. 4^6 added 4 times equals: 1

(a) 4^{24} **(b)** 16^6 **(c)** 4^{10} **(d)** 4^7

8. If $5^{2x+3} = 5^{13}$, the value of x is: 1

(a) 10 **(b)** 5 **(c)** 6 **(d)** 8

9. **Assertion (A):** $3^0 + 2^0 = 2$. 1

Reason (R): $a^0 = 0$ for every non-zero number a.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $2^3 \times 3^3 = 5^3$. 1

Reason (R): $a^m \times b^m = (ab)^m$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write $(2.5 \times 10^6) + (5 \times 10^5)$ in scientific notation. 2

12. Find the ones digit of 4^{61}. 2

13. Find x if $5^{2x-3} = 125$. 2

14. Write $(7 \times 10^{-2}) + (3 \times 10^{-3})$ in scientific notation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A sheet of paper 0.05 mm thick is folded in half 12 times (assume this is possible). Find the thickness of the folded paper in mm and in cm. 3

16. Light travels 3×10^8 m in one second. How far does it travel in 8 minutes? Give the answer in scientific notation. 3

17. Find the value of $\dfrac{5^4 + 5^3}{5^2 \times (5+1)}$. 3

OR

A sheet of paper 0.4 mm thick is folded in half 6 times (assume this is possible). Find the thickness of the folded paper in mm and in cm. 3

18. Light travels 3×10^8 m in one second. How far does it travel in 12 minutes? Give the answer in scientific notation. 3

OR

Find the value of $\dfrac{4^7 + 4^6}{4^5 \times (4+1)}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A biology student in Pune starts a culture of 5 bacteria that double in number every hour. She records the number after n hours as 5×2^n. 5

(i) How many bacteria are there after 1 hour?
(ii) How many are there after 6 hours?
(iii) After how many whole hours does the count first exceed 1000?
(iv) Using laws of exponents, find the ratio of the count after 8 hours to the count after 4 hours.

20. A culture has 3 bacteria and the number becomes 5 times every hour. Write the number after 3 hours as an expression with a power, find its value, and say how many times the original number it is. 5

OR

Body A has mass 1.8×10^{15} kg and body B has mass 6×10^{12} kg. Which is heavier? How many times as heavy is it? Find the total mass in scientific notation. 5

END OF QUESTION PAPER — **Total: 40 Marks**